

3. Oracle jet components

Note: The practices of the chapter 6,7,8,9 are covered progressively in this chapter. After this chapter, the next chapter will be chapter 10.

3.1. Creating and Adding a Simple Form in Oracle JET using NavDrawer Template

Introduction: In this activity, you'll learn how to integrate a **basic form component** inside a custom route (Form Practice). This hands-on exercise will introduce you to Oracle JET's layout, routing, and form UI components.

Purpose:

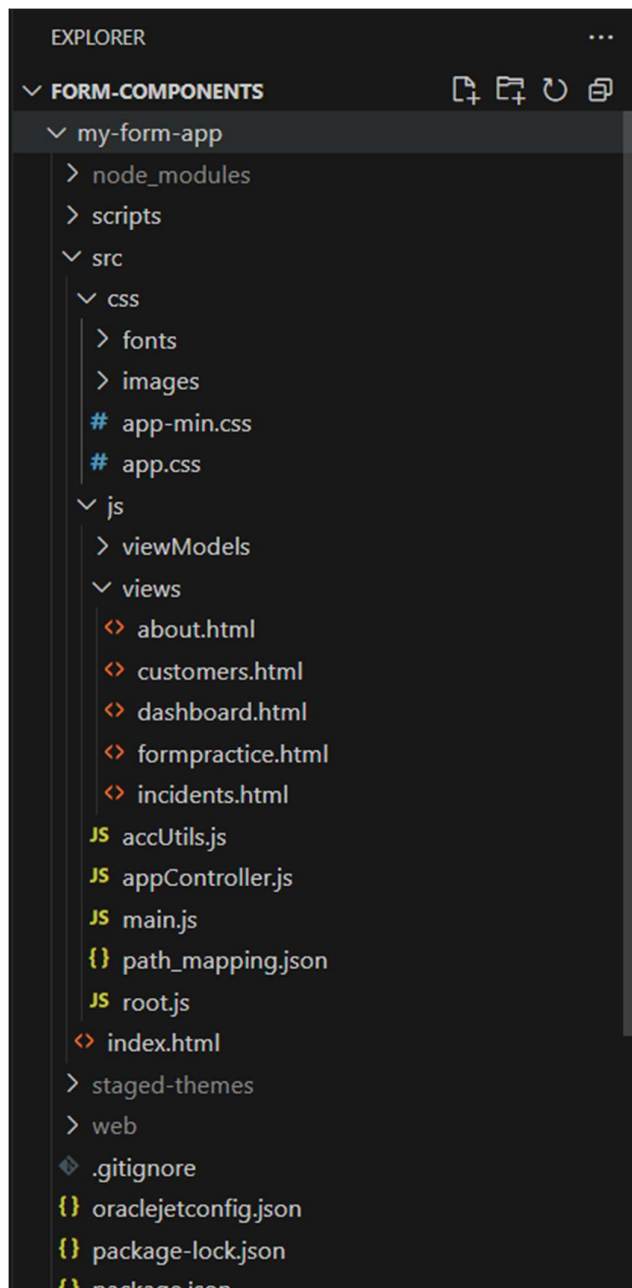
- To understand the structure of a JET application using the NavDrawer template.
- To practice modifying navigation routes.
- To integrate `<oj-form-layout>` and form components like `<oj-input-text>` in a new view.
- To prepare the foundation for more complex form activities (e.g., validation, submission).

Pre-requisites:

Before starting this activity, ensure the following:

- Oracle JET CLI is installed (`npm install -g @oracle/ojet-cli`)
- Oracle JET app is created using NavDrawer template
- Node.js and npm are properly configured
- Editor (like VS Code) is ready
- You've tested the base app with `ojet serve`

Project structure:



Tasks:

1. Create the application “my-form -app” through navdrawer template and move to the project directory by using change directory command.

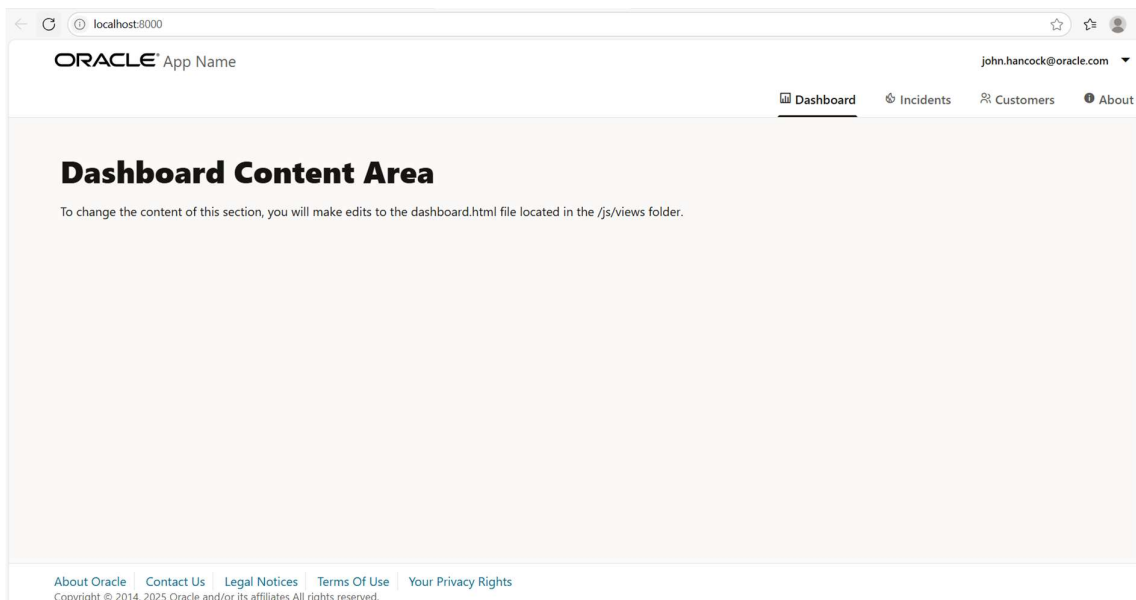
```
PS D:\Courses\Oracle Jet\ojet labs\form-components> ojet create my-form-app --template=navdrawer ✓
Processing template: navdrawer
Your app structure is generated. Continuing with library install.
oraclejetconfig.json file exists
Running after_app_create hook.
Your app is ready! Change to your new app directory 'my-form-app' and try 'ojet build' and 'ojet serve'.
PS D:\Courses\Oracle Jet\ojet labs\form-components> cd my-form-app ✓
PS D:\Courses\Oracle Jet\ojet labs\form-components\my-form-app> ojet build web
Cleaning staging path.
Running before_build hook.
```

2. Run the command “ Ojet build web” to **build your Oracle JET application** for the **web platform** in a production-ready format.

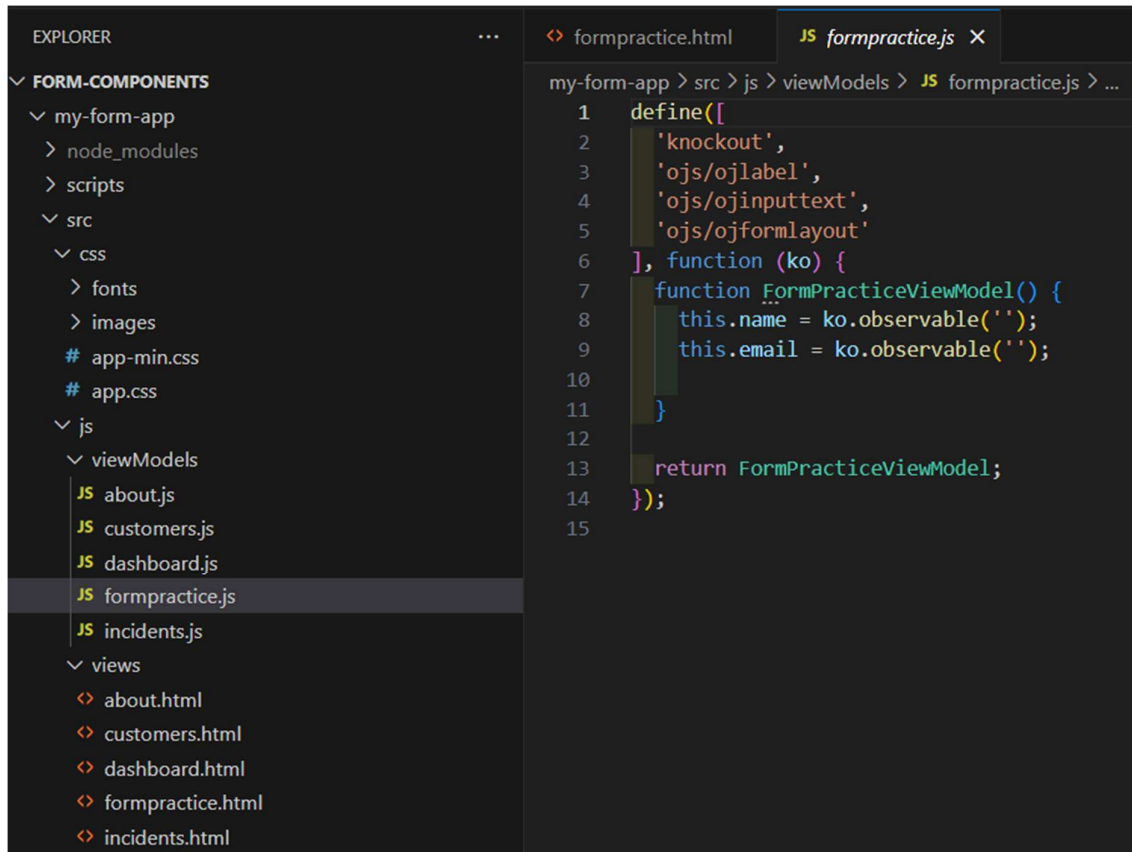
```
PS D:\Courses\Oracle Jet\ojet labs\form-components\my-form-app> ojet build web
Cleaning staging path.
Running before_build hook.
Copy files to staging directory.
Copy finished.
Copy library files to staging directory.
Copy finished.
Copy reference components to staging directory.
```

3. Run the command “ ojet serve” .It Runs the Oracle JET app in a local development server with live reload, so you can test your changes instantly in the browser.

```
Success: Build finished.
PS D:\Courses\Oracle Jet\ojet labs\form-components\my-form-app> ojet serve
Warning: Command is missing platform. Default to web.
Build: true
BuildType: dev
Destination: undefined
Destination target: undefined
Livereload: true
Watch files: true
```

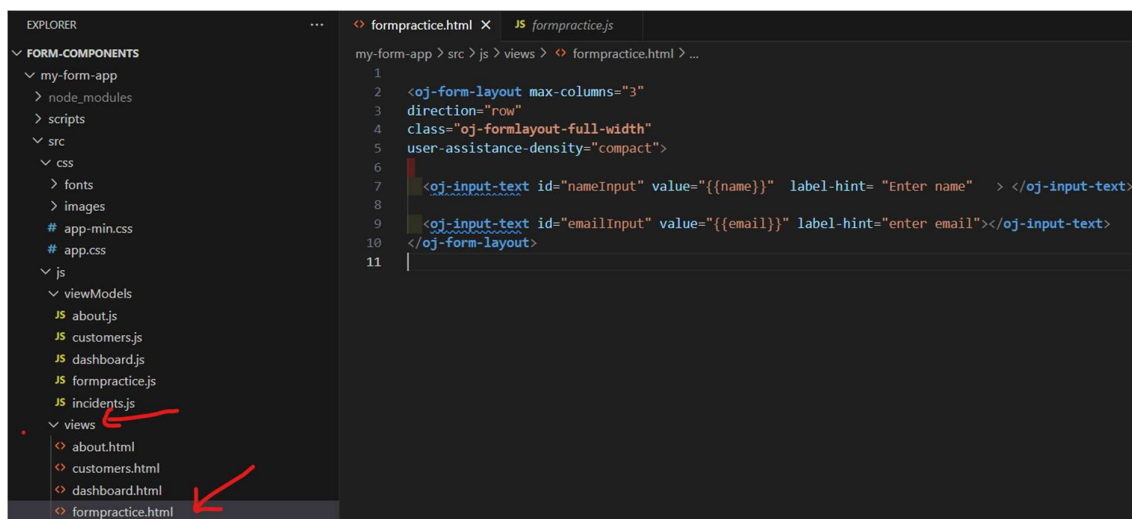


4. Now add the formpractice.js file and formpractice.html file under the viewModels and Views directory respectively.



```
1  define([
2    'knockout',
3    'ojs/ojlabel',
4    'ojs/ojinputtext',
5    'ojs/ojformlayout'
6  ], function (ko) {
7    function FormPracticeViewModel() {
8      this.name = ko.observable('');
9      this.email = ko.observable('');
10    }
11
12    return FormPracticeViewModel;
13  });
```

And formpractice.html



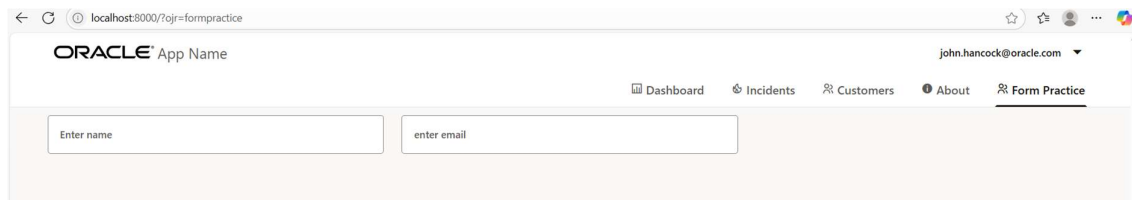
```
1  <oj-form-layout max-columns="3"
2    direction="row"
3    class="oj-formlayout-full-width"
4    user-assistance-density="compact">
5
6
7    <oj-input-text id="nameInput" value="{{name}}" label-hint="Enter name" > </oj-input-text>
8
9    <oj-input-text id="emailInput" value="{{email}}" label-hint="enter email"> </oj-input-text>
10 </oj-form-layout>
11
```

5. Add New Route for Form Practice. This adds the "Form Practice" entry in the navigation drawer.

```
{ path: 'formpractice', detail: { label: 'Form Practice', iconClass: 'oj-ux-ico-contact-group' } },
```

```
let navData = [  
  { path: '', redirect: 'dashboard' },  
  { path: 'dashboard', detail: { label: 'Dashboard', iconClass: 'oj-ux-ico-bar-chart' } },  
  { path: 'incidents', detail: { label: 'Incidents', iconClass: 'oj-ux-ico-fire' } },  
  { path: 'customers', detail: { label: 'Customers', iconClass: 'oj-ux-ico-contact-group' } },  
  { path: 'about', detail: { label: 'About', iconClass: 'oj-ux-ico-information-s' } },  
  { path: 'formpractice', detail: { label: 'Form Practice', iconClass: 'oj-ux-ico-contact-group' } },  
];
```

6. Restart the app if needed: `ojet serve`
7. Open your browser and navigate to `http://localhost:8000`
8. Click on **Form Practice** from the side NavDrawer
9. You should see two input fields: **Name** and **Email**



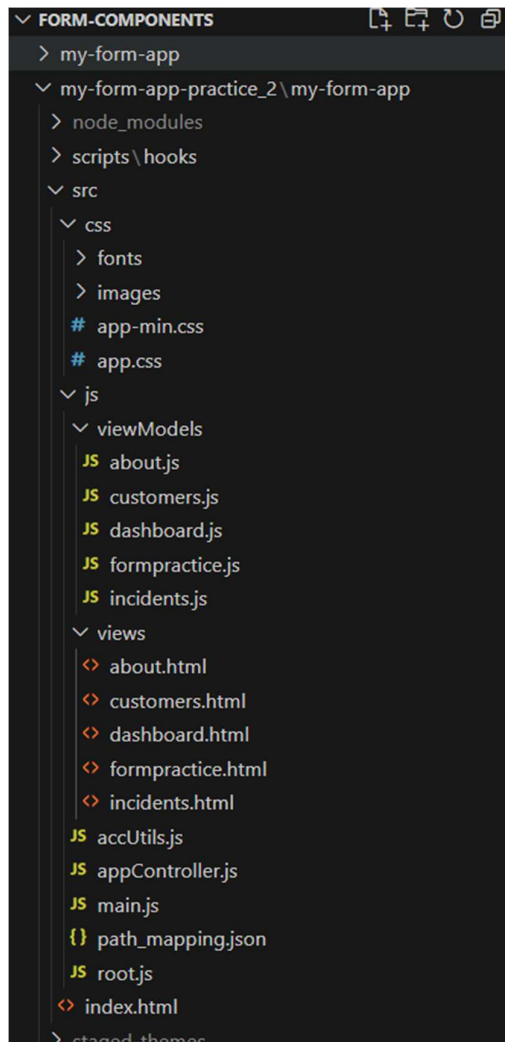
3.2. Oracle JET Form with Text Inputs and Dropdown Using NavDrawer Template

Introduction : In this activity, you'll expand your Oracle JET SPA by adding a **form with text inputs and a dropdown** to the “Form Practice” route. You’ll use `oj-input-text` and `oj-select-single` components arranged within a `oj-form-layout`.

Pre-requisite :

- **Practice 1 (Basic Form with Name and Email)** is already completed.
- You have already added a working `formpractice.js` and `formpractice.html` file under `/js/viewModels/` and `/js/views/`.

Project structure:



Note : For each practice, we are progressively including new functionalities while retaining the previous ones. This ensures that earlier features continue to work alongside the newly added ones.

Tasks :

1. Create a folder my-form-app-practice_2 and create a project named "my-form-app"

ojet create my-form-app --template=navdrawer

```
Install the latest POWERShell for new features and improvements! https://aka.ms/PSWindows  
  
PS D:\Courses\Oracle Jet\ojet labs\form-components\my-form-app-practice_2> ojet create my-form-app --template=navdrawer  
Processing template: navdrawer  
Your app structure is generated. Continuing with library install.  
oraclejetconfig.json file exists  
Running after_app_create hook.  
Your app is ready! Change to your new app directory 'my-form-app' and try 'ojet build' and 'ojet serve'.
```

2. Change directory to my-form-app project and then build and serve the project. This will create a empty project with predefined UI

```
PS D:\Courses\Oracle Jet\ojet labs\form-components\my-form-app-practice_2> cd my-form-app  
PS D:\Courses\Oracle Jet\ojet labs\form-components\my-form-app-practice_2\my-form-app> ojet build web  
Cleaning staging path.  
Running before_build hook.  
Copy files to staging directory.  
Copy finished.  
Copy library files to staging directory.  
Copy finished.  
Copy reference components to staging directory.
```

3. Run ojet serve to see the live review of the page

```
Success: Build finished.  
PS D:\Courses\Oracle Jet\ojet labs\form-components\my-form-app-practice_2\my-form-app> ojet serve  
Warning: Command is missing platform. Default to web.  
Build: true  
BuildType: dev  
Destination: undefined  
Destination target: undefined
```

4. Now create formpractice.js file under viewmodels directory , which is under js folder

```

JS formpractice.js X
my-form-app-practice_2 > my-form-app > src > js > viewModels > JS formpractice.js > ...
1  define([
2    'knockout',
3    'ojs/ojlabel',
4    'ojs/ojinputtext',
5    'ojs/ojformlayout',
6    'ojs/ojselectsingle',
7    'ojs/ojarraydataprovider'
8  ], function (ko, ojlabel, ojinputtext, ojformlayout, ojselectsingle, ArrayDataProvider) {
9
10     function FormPracticeViewModel() {
11       this.name = ko.observable('');
12       this.email = ko.observable('');
13
14       // Practice 2: Country Dropdown
15       this.selectedCountry = ko.observable('');
16       this.countryList = [
17         { value: 'IN', label: 'India' },
18         { value: 'US', label: 'United States' },
19         { value: 'UK', label: 'United Kingdom' },
20         { value: 'CA', label: 'Canada' },
21         { value: 'AU', label: 'Australia' }
22       ];
23       this.countryDataProvider = new ArrayDataProvider(this.countryList, { keyAttributes: 'value' });
24     }
25
26     return FormPracticeViewModel;
27 });

```

5. Create formpractice.html file under views folder

```

<> formpractice.html X
my-form-app-practice_2 > my-form-app > src > js > views > <> form
1
2 <div>
3 <oj-form-layout max-columns="3"
4   direction="row"
5   class="oj-formlayout-full-width"
6   user-assistance-density="compact">
7
8   <!-- Practice 1 Fields -->
9   <oj-input-text
10    id="nameInput"
11    value="{{name}}"
12    label-hint="Enter name"
13    required="true">
14   </oj-input-text>
15
16   <oj-input-text
17    id="emailInput"
18    value="{{email}}"
19    label-hint="Enter email"
20    required="true">
21   </oj-input-text>
22
23   <!-- Practice 2: Country Dropdown -->
24   <oj-select-single
25    id="countrySelect"
26    label-hint="Select Country"
27    data="[[countryDataProvider]]"
28    item-text="label"
29    value="{{selectedCountry}}"
30    required="true">
31   </oj-select-single>
32
33 </oj-form-layout>
34

```

6. Edit appcontroller.js file (like we did in previous practice)

```

let navData = [
  { path: '', redirect: 'dashboard' },
  { path: 'dashboard', detail: { label: 'Dashboard', iconClass: 'oj-ux-ico-bar-chart' } },
  { path: 'incidents', detail: { label: 'Incidents', iconClass: 'oj-ux-ico-fire' } },
  { path: 'customers', detail: { label: 'Customers', iconClass: 'oj-ux-ico-contact-group' } },
  { path: 'about', detail: { label: 'About', iconClass: 'oj-ux-ico-information-s' } },
  { path: 'formpractice', detail: { label: 'Form Practice', iconClass: 'oj-ux-ico-contact-group' } },
];

```

7. Run ojet serve on command terminal
8. Navigate to the Form Practice page in the browser
9. Enter Name, Email, and select a Country

Upto this practice, you have created two textboxes and one drop down list for the form